

Sayed M. Asif

917-609-5130 | sayed.asif36@gmail.com | linkedin.com/in/sayedmasif

EDUCATION

University at Buffalo , B.S. in Mechanical Engineering, GPA: 3.2	Buffalo, NY
University at Buffalo , M.S. in Mechanical Engineering, GPA: 3.5	Buffalo, NY
• Engineer in Training (EIT): Passed the Fundamentals of Engineering (FE) exam • Awards: AIAA 2022 Region 1 Student Conference Winner, Team Category	

TECHNICAL SKILLS

Programming Languages: MATLAB, C++, Python
Software: AutoCAD, Autodesk Inventor, ANSYS LS-DYNA, Creo, SolidWorks, Microsoft Office Suite (Excel, Word, PowerPoint, Visual Studio, Teams), Arduino IDE
Soft Skills: Project Management, Technical Documentation, Cross-Functional Collaboration, Customer Service
Engineering Competencies: CAD Modeling, Finite Element Analysis, Structural Analysis, Precision Motion Control Systems, Geometric Dimensioning and Tolerance, Rapid Prototyping, Electromechanical Systems Integration, Thermal Management, Fluid Mechanics, Technical Writing, Statistical Analysis, Machine Design Principles, Motion Control Algorithms, System-Level Troubleshooting, Precision Assembly Processes, Soft Robotics, Sensor Integration

WORK EXPERIENCE

Exponent <i>Research Assistant</i>	Sep. 2025 - Oct. 2025 New York, NY
• Assisted with collection and processing of 800 hours of audio-visual data across 3 boroughs of NYC for AI research. • Processed data through video editing software to remove sensitive information and improve data reliability. • Coordinated with cross-functional team of 15 people to assemble and deliver data collection equipment to research participants, retrieve data collection equipment, and also store and process data.	
CRASH Lab <i>Research Assistant</i>	May 2021 - Jan. 2025 Buffalo, NY
• Engineered motor control systems according to precise technical specifications for aerospace-based robotic systems. • Developed and optimized control algorithms with MATLAB and C++, improving accuracy by 10%. • Created electromechanical systems, achieving 20% weight reduction while meeting aerospace performance criteria.	
University at Buffalo <i>Graduate Student Assistant</i>	Aug. 2022 - Jun. 2023 Buffalo, NY
• Mentored students on CAD and FEA tools, accelerating design iteration by 3 weeks through customized tutorials. • Guided students on ultrasonic sensor optimization, improving accuracy by 25% for automation-related applications. • Supported senior design course, streamlining coursework and grading processes, reducing turnaround time by 30%.	

PROJECTS

FEA of Aircraft Wing Structures <i>Autodesk Inventor, ANSYS LS-DYNA, Structural Analysis</i>	Dec. 2022
• Performed finite element analysis using ANSYS LS-DYNA, achieving specifications within 0.5%, validating precision mechanical system integrity. • Prepared technical drawings with Autodesk Inventor, reducing material usage by 8% for aerospace components. • Delivered analysis and technical reports 2 weeks ahead of schedule, supporting design validation and optimization.	
Compliance-Based Bio-mimetic Wings <i>Autodesk Inventor, SolidWorks, Rapid Prototyping</i>	May 2024
• Modeled a motor-driven soft robotic component using Autodesk Inventor and SolidWorks, producing technical drawings and improving motion control accuracy by 30%. • Developed a PID control system for actuation, reducing errors by 12%, enhancing reliability for robotic systems. • Fabricated and tested a prototype, completing validation 10 days faster than planned for aerospace applications.	
Portable Modular Car Shelter <i>AutoCAD, MATLAB, Mechanical Design</i>	May 2022
• Designed a portable car shelter that meets customer and technical specifications using Solidworks, creating detailed layouts and BOMs, reducing assembly time by 1 hour. • Validated designs using MATLAB, achieving 95% correlation with tests, optimizing precision and reliability. • Collaborated in a team of 5 to deliver the project within budget, saving \$100 on material costs.	
Control Algorithm for Autonomous Rover <i>Arduino, Control Systems, Automation Integration</i>	May 2024
• Designed a motion control algorithm for an autonomous rover using Arduino, improving accuracy by 20% in dynamic environments. • Implemented a PID control algorithm, reducing response time by 0.5 s, enhancing automation system performance. • Integrated mechanical, electrical, and software components, achieving 90% reliability during testing.	